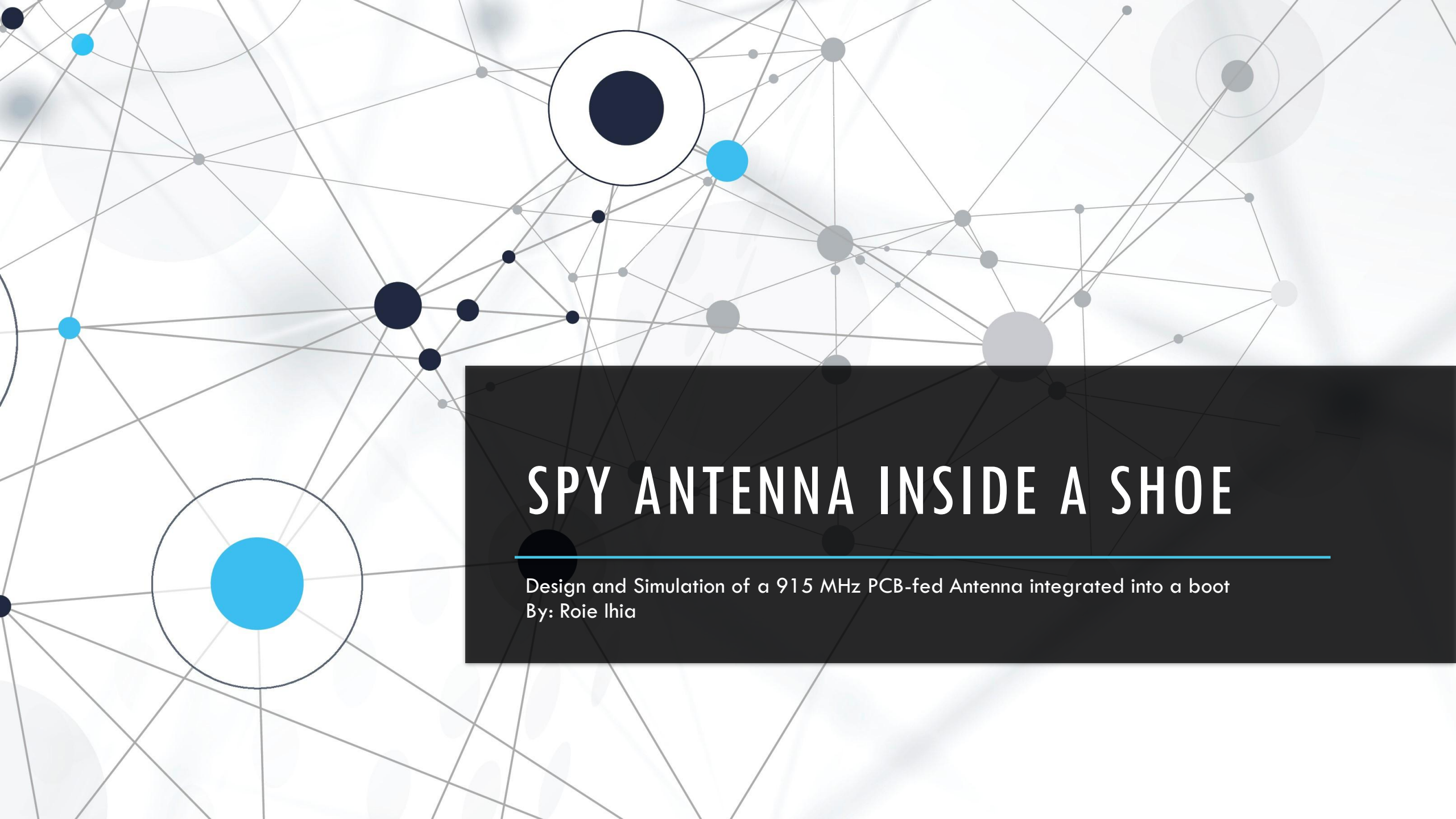




SPY ANTENNA INSIDE A SHOE

Design and Simulation of a 915 MHz PCB-fed Antenna integrated into a boot
By: Roie Ihia

SPY ANTENNA INSIDE A SHOE



ENGINEERING
CONSIDERATIONS AND
CHOICES



MODELING IN FREE SPACE



ANTENNA INSIDE A SHOE
DEMO

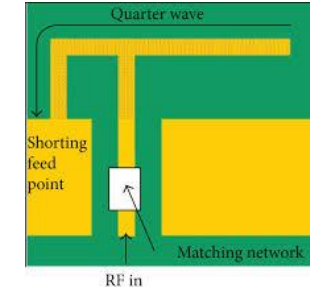
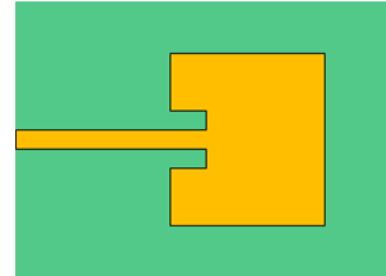
ENGINEERING CONSIDERATIONS & TOPOLOGY CHOICES

1. Size Constraints:

- a) Small and complex area to work on
- b) $f = 915 \text{ MHz} \rightarrow \lambda \cong 32 \text{ cm}$

2. Lossy Environment:

- a) The human foot is a high-loss dielectric ($40 \leq \epsilon_{r_{body}} \leq 45$) it absorbs RF energy and shifts the resonant frequency downward.
- b) Proximity to the Ground – Reflections
- c) Robustness



Patch Antenna

- Pros: Excellent directivity away from the foot
- Cons: A standard half-wave patch at 915 MHz requires an area of roughly 8×8

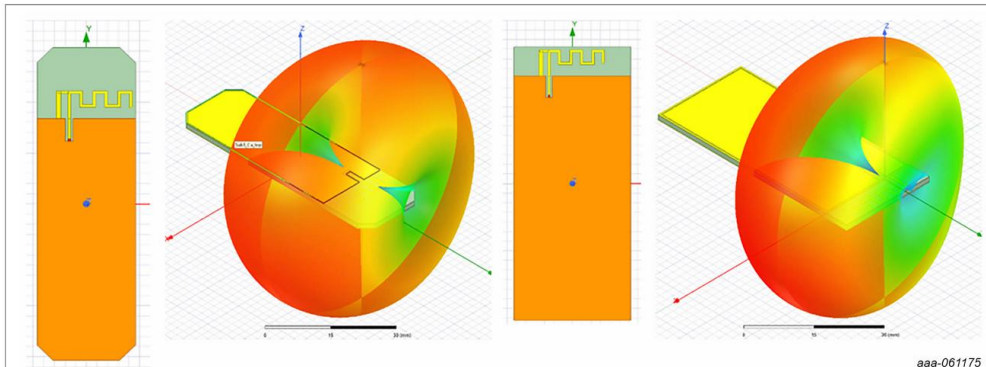
PIFA Antenna (Printed Inverted F Antenna)

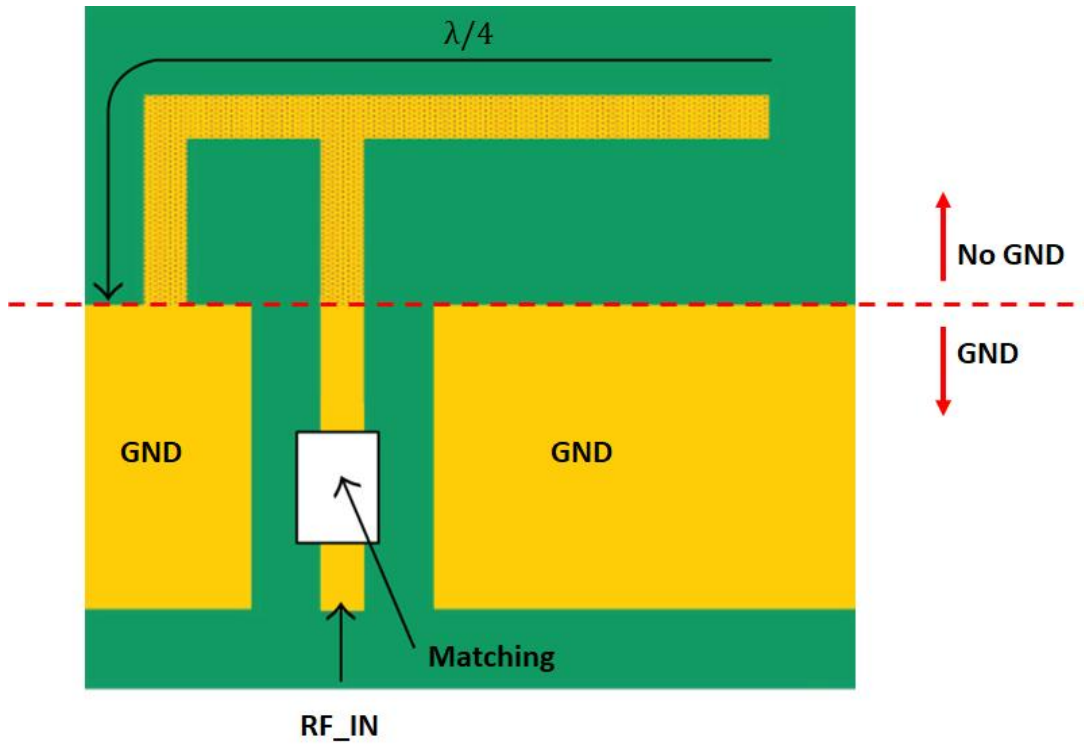
- Pros: This design resonates at $\frac{\lambda}{4}$ meaning an small form factor
- Cons: Narrower bandwidth and lower peak directivity compared to a full patch

ENGINEERING CONSIDERATIONS AND CHOICES

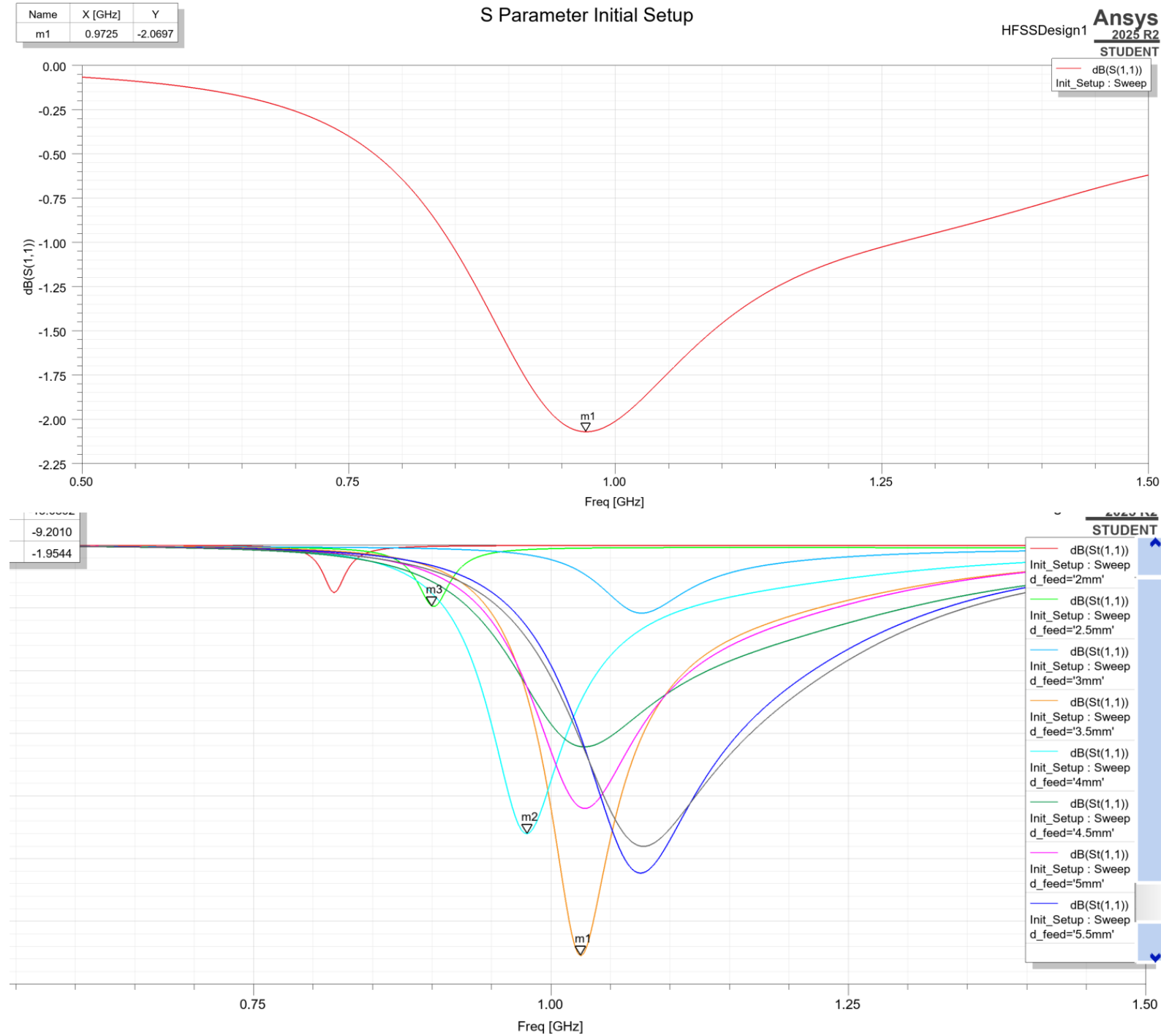
Location Selection: Balancing RF and Mechanics:

- Option A- The Shoe Sole: Extreme proximity to the ground creates severe propagation fast-fading and multipath interference.
- Option B- The Top of the Foot : This area flexes heavily with the kinetic function of the foot, which would mechanically deform the PCB antenna.
- Option C- The Heel Counter (Back of the Boot): The heel counter is rigid, preventing mechanical detuning. Crucially, placing the antenna here aligns the principle radiation to point in an off-body direction.

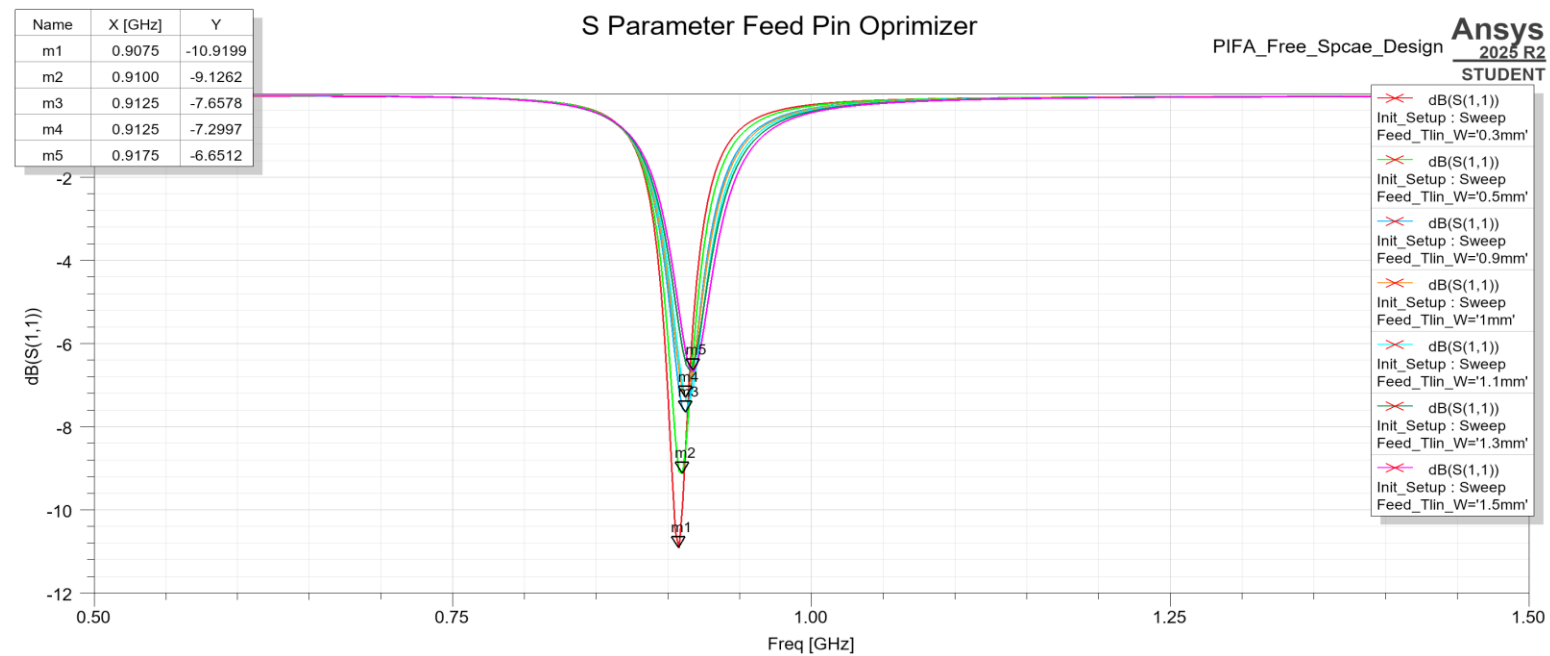
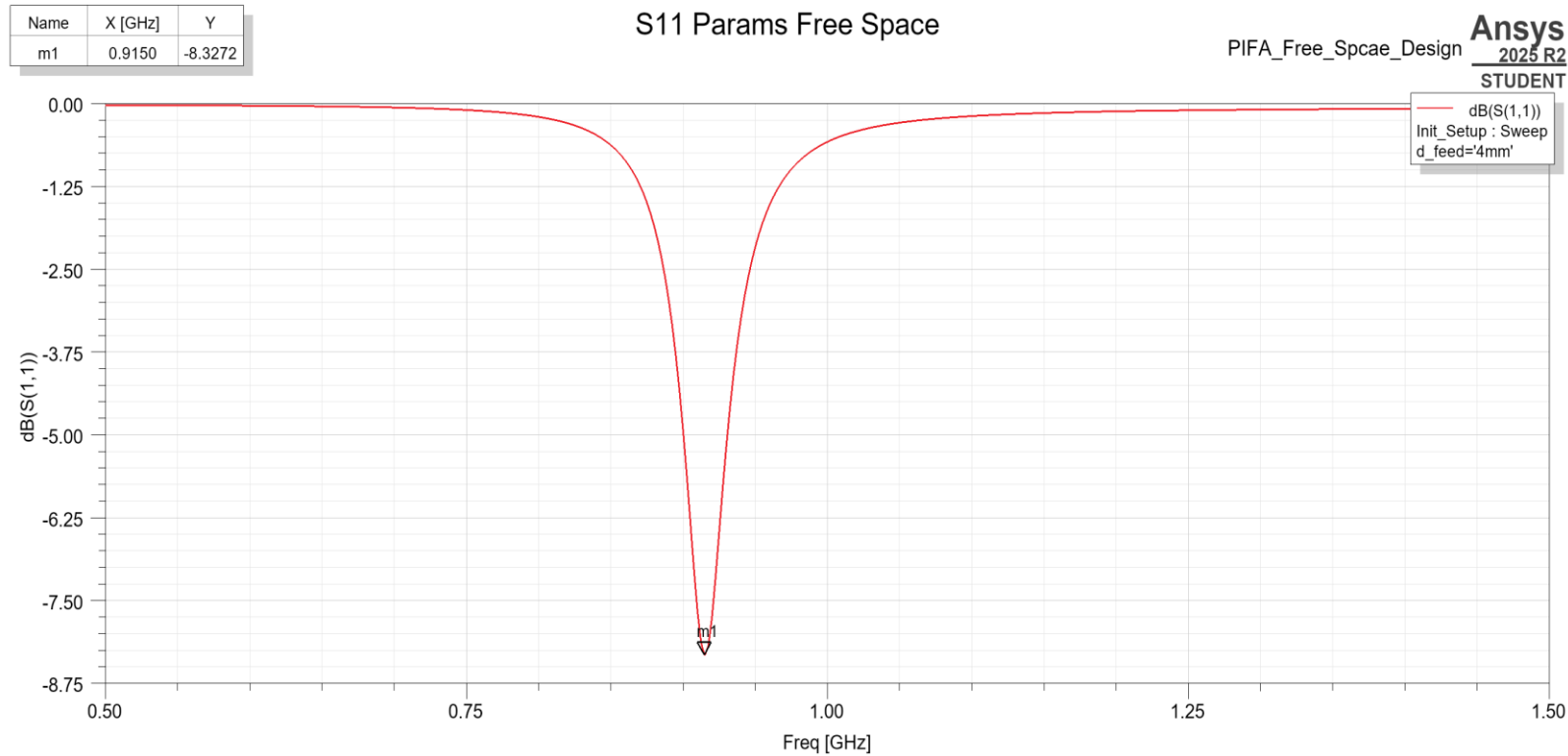


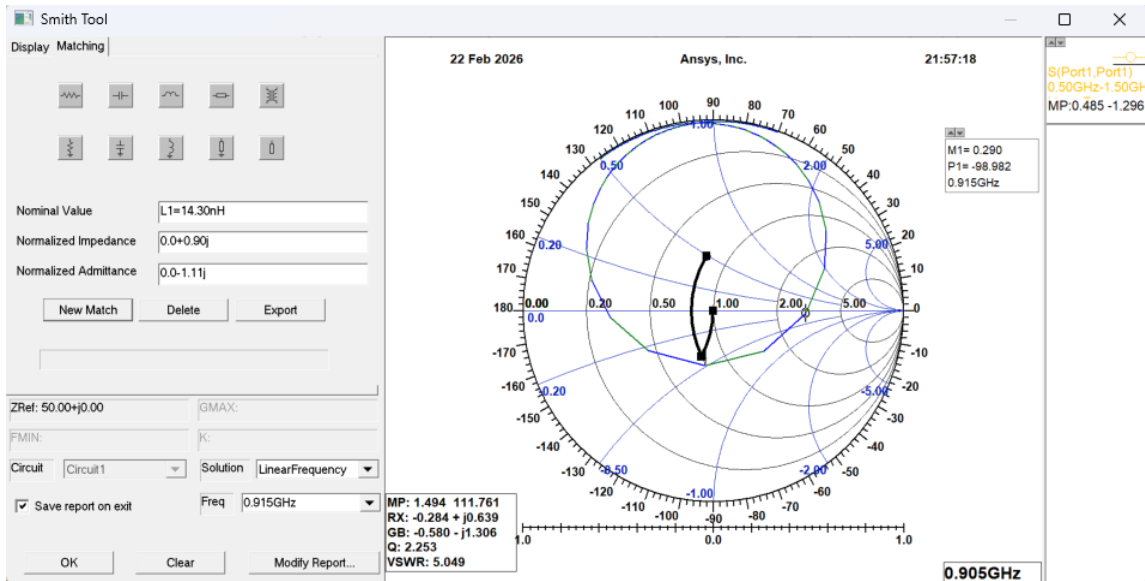


MODELING IN FREE SPACE

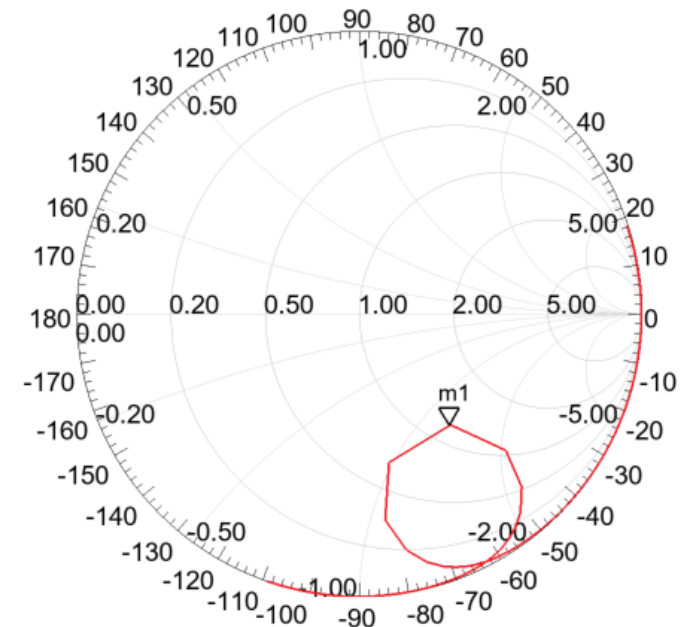


MODELING IN FREE SPACE

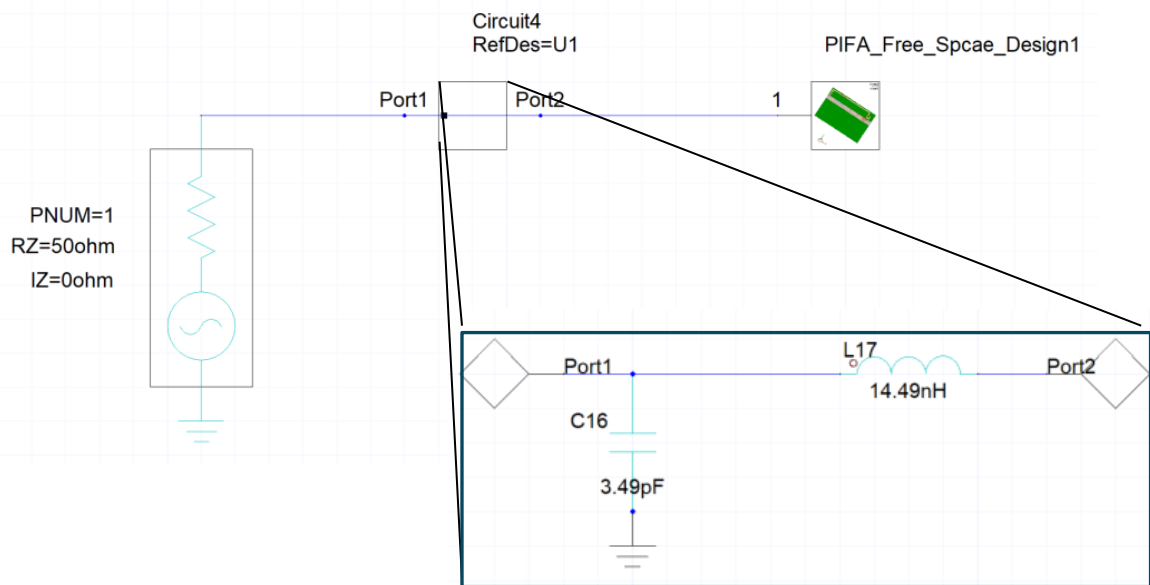


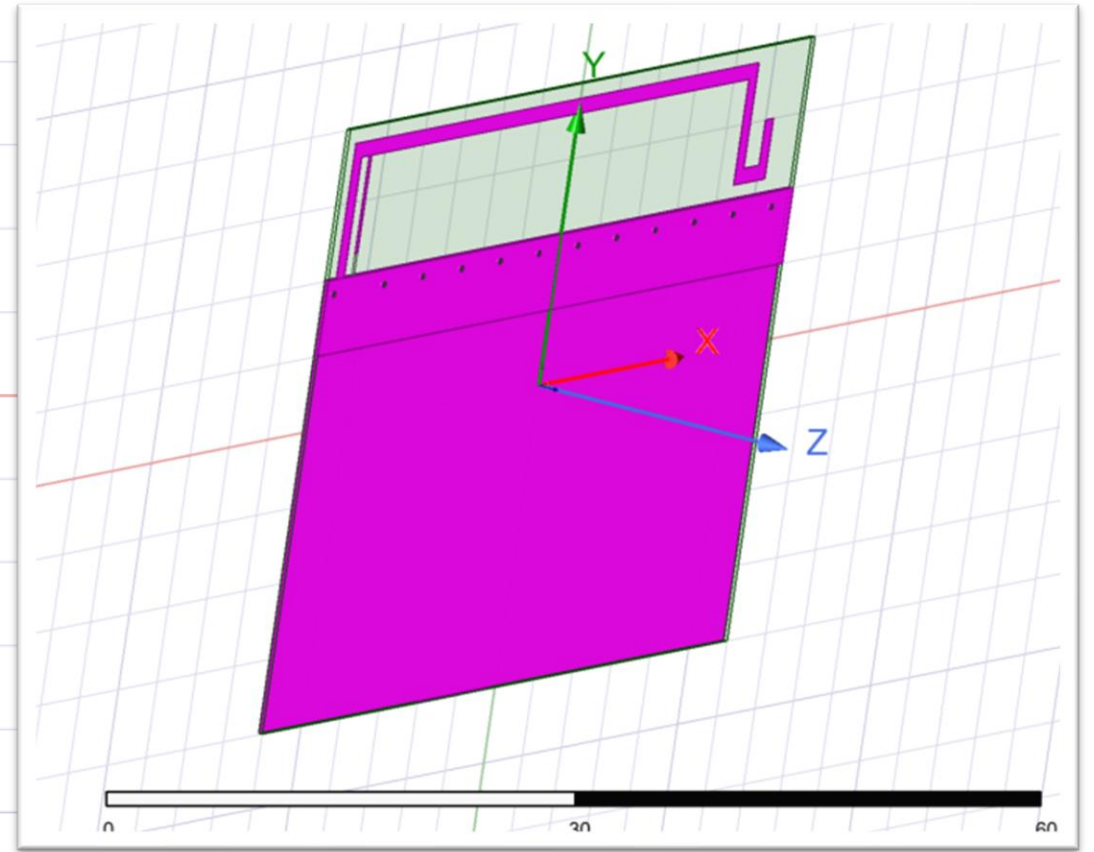
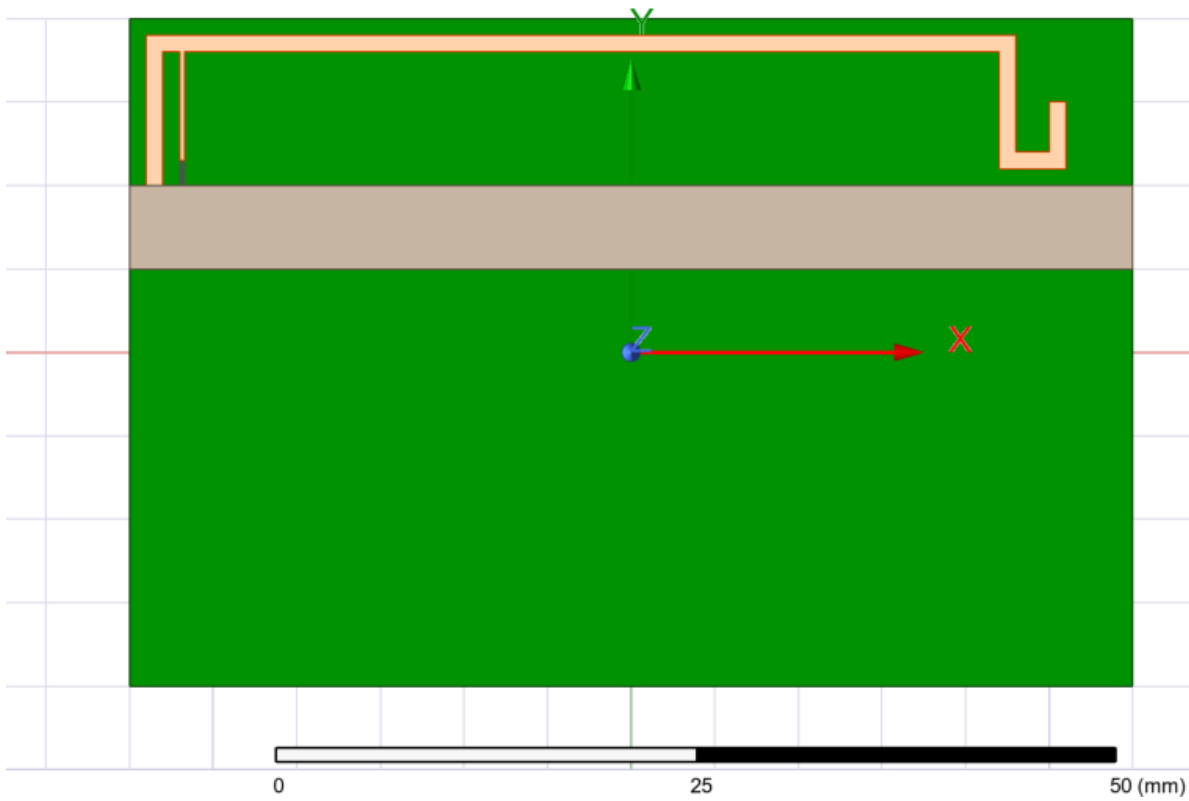


S Parameter Matched Smith

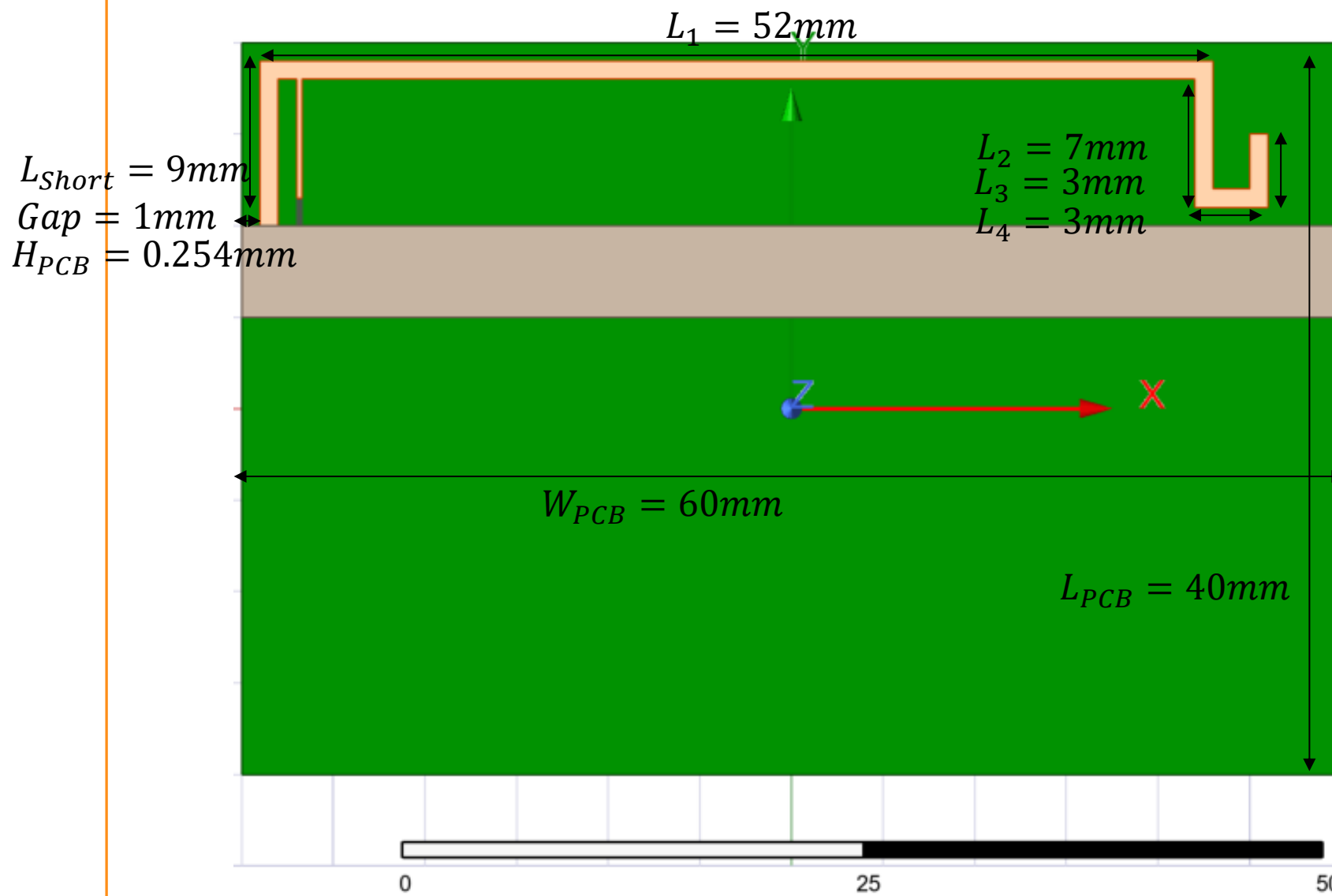


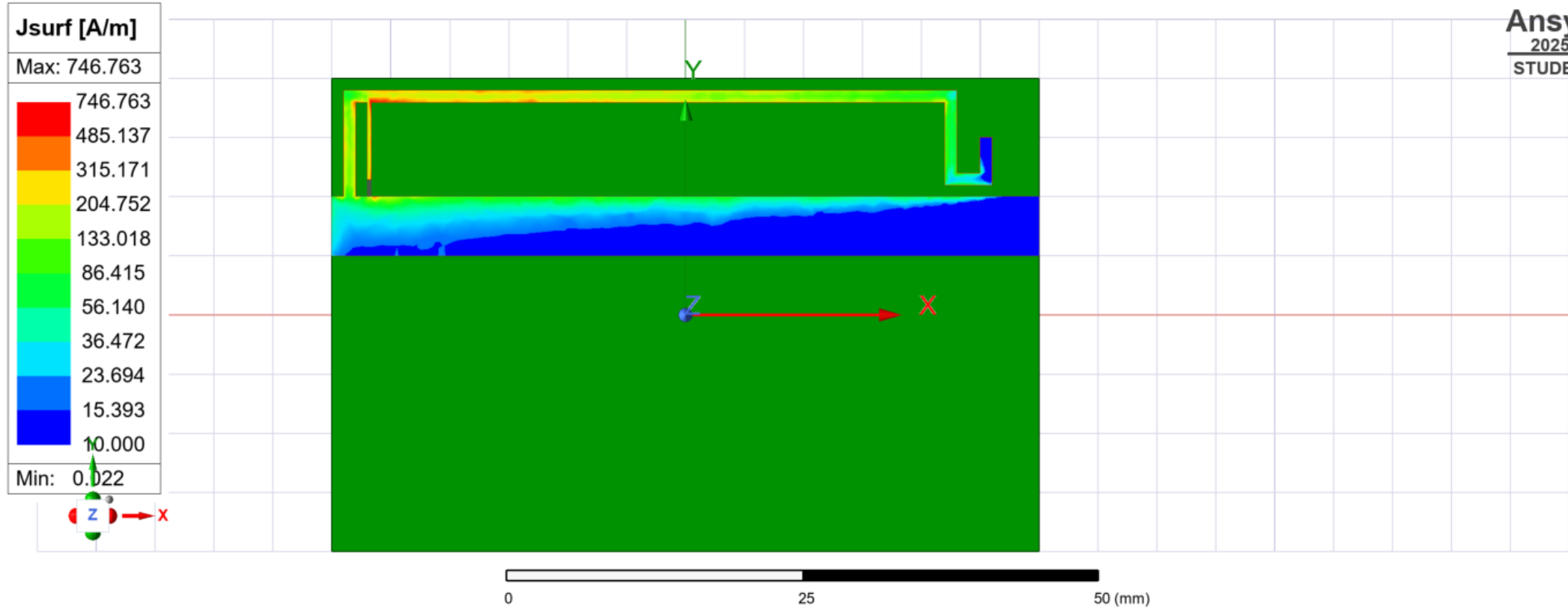
Name	Freq [MHz]	Ang	Mag	RX
m1	915	-50.6340	0.5073	1.2099 - 1.2778i





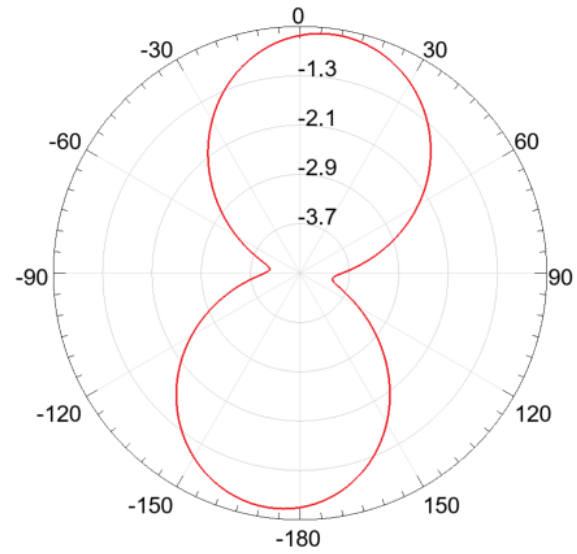
MODELING IN FREE SPACE — FINAL
DESIGN



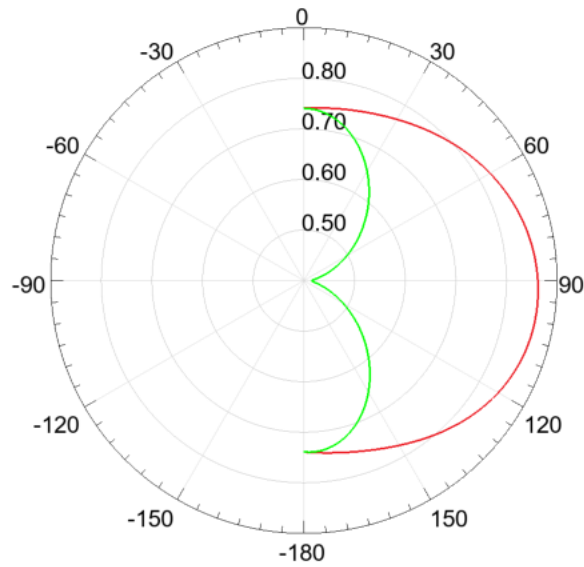


SURFACE CURRENT PLOT

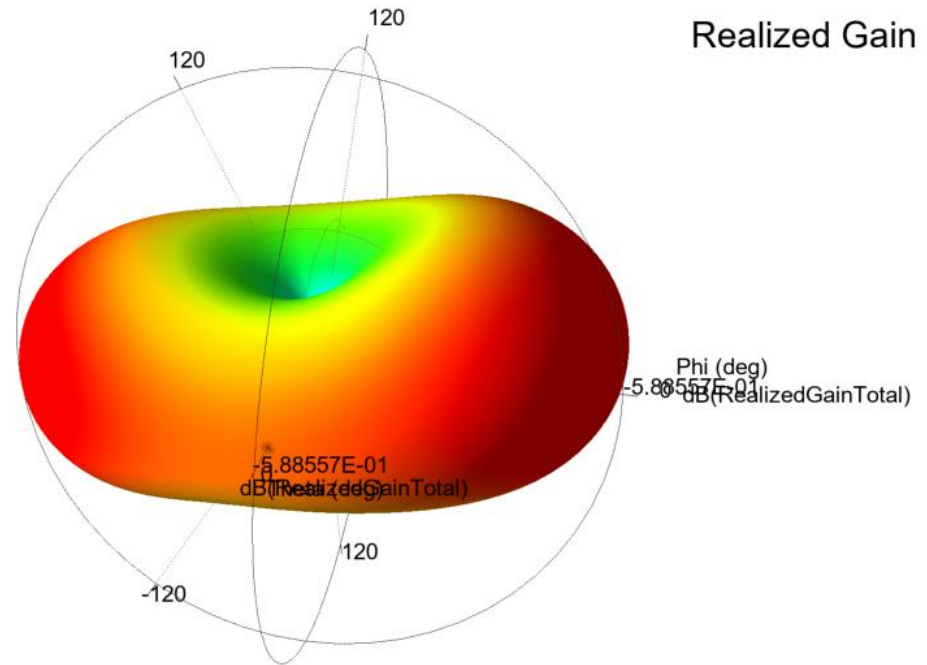
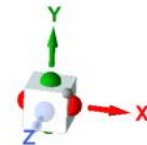
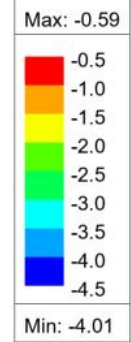
Relized Gain AZ Cut (XY Plain)



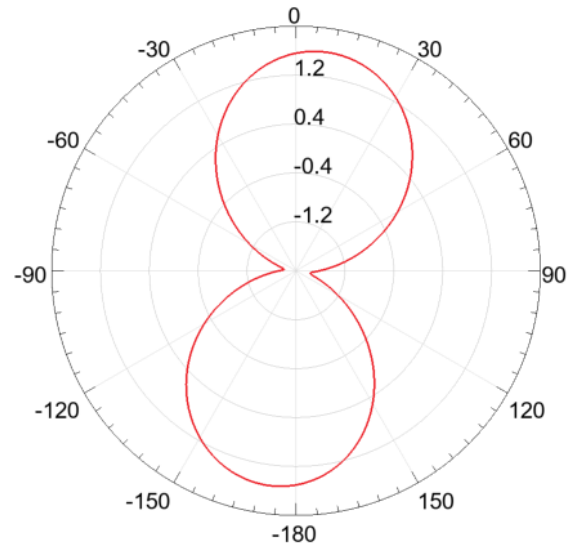
Realized Gain EL Cut (XZ, YZ)



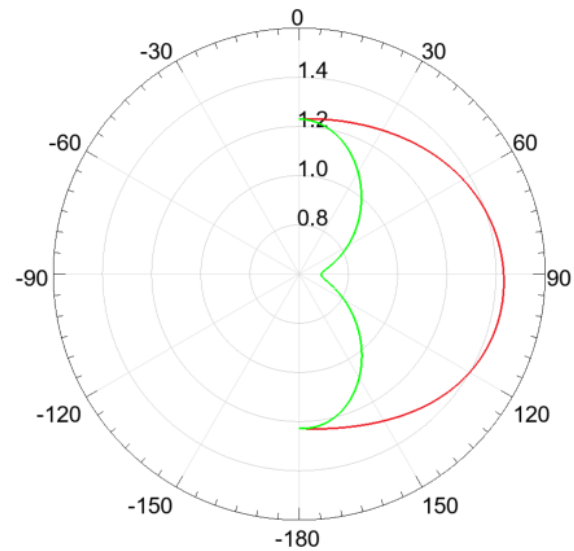
Ansys Inc.



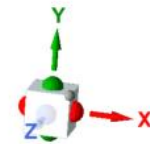
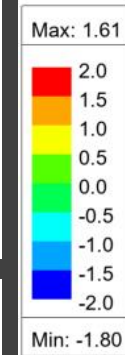
Directivity AZ Cut (XY Plain)



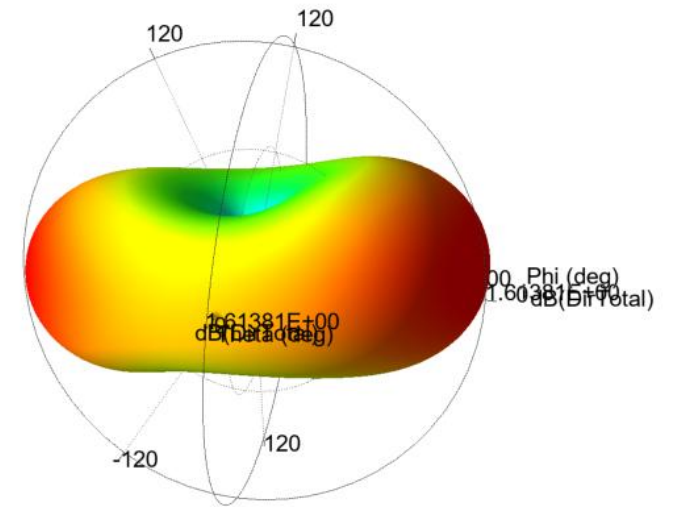
Directivity EL Cut (XZ, YZ)

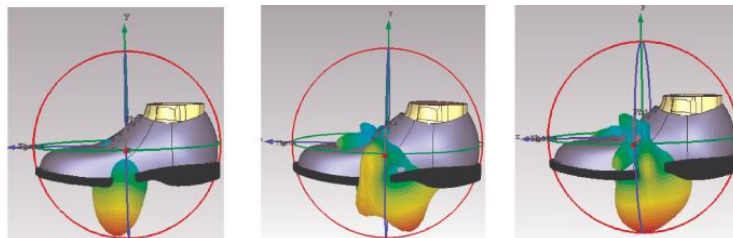


Ansys Inc.



Directivity



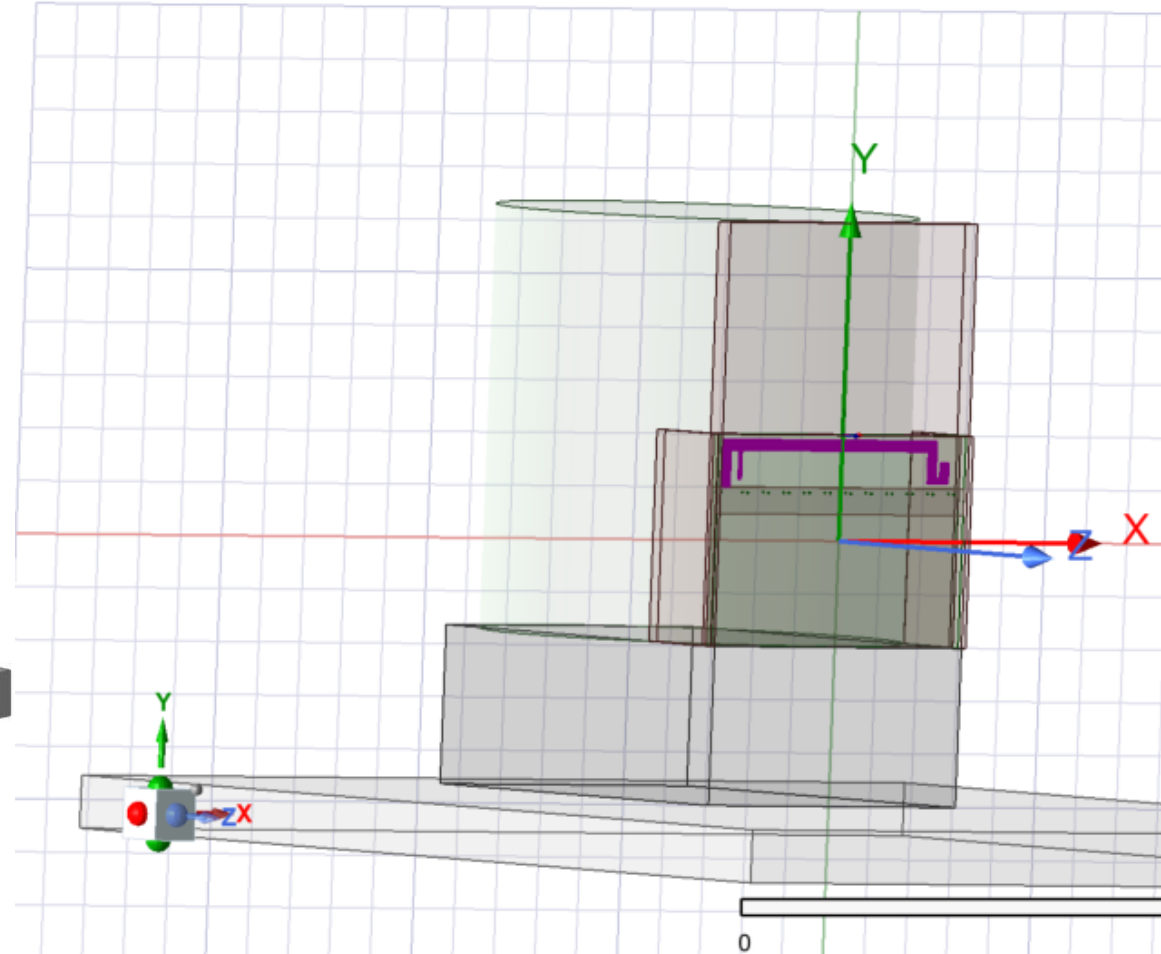
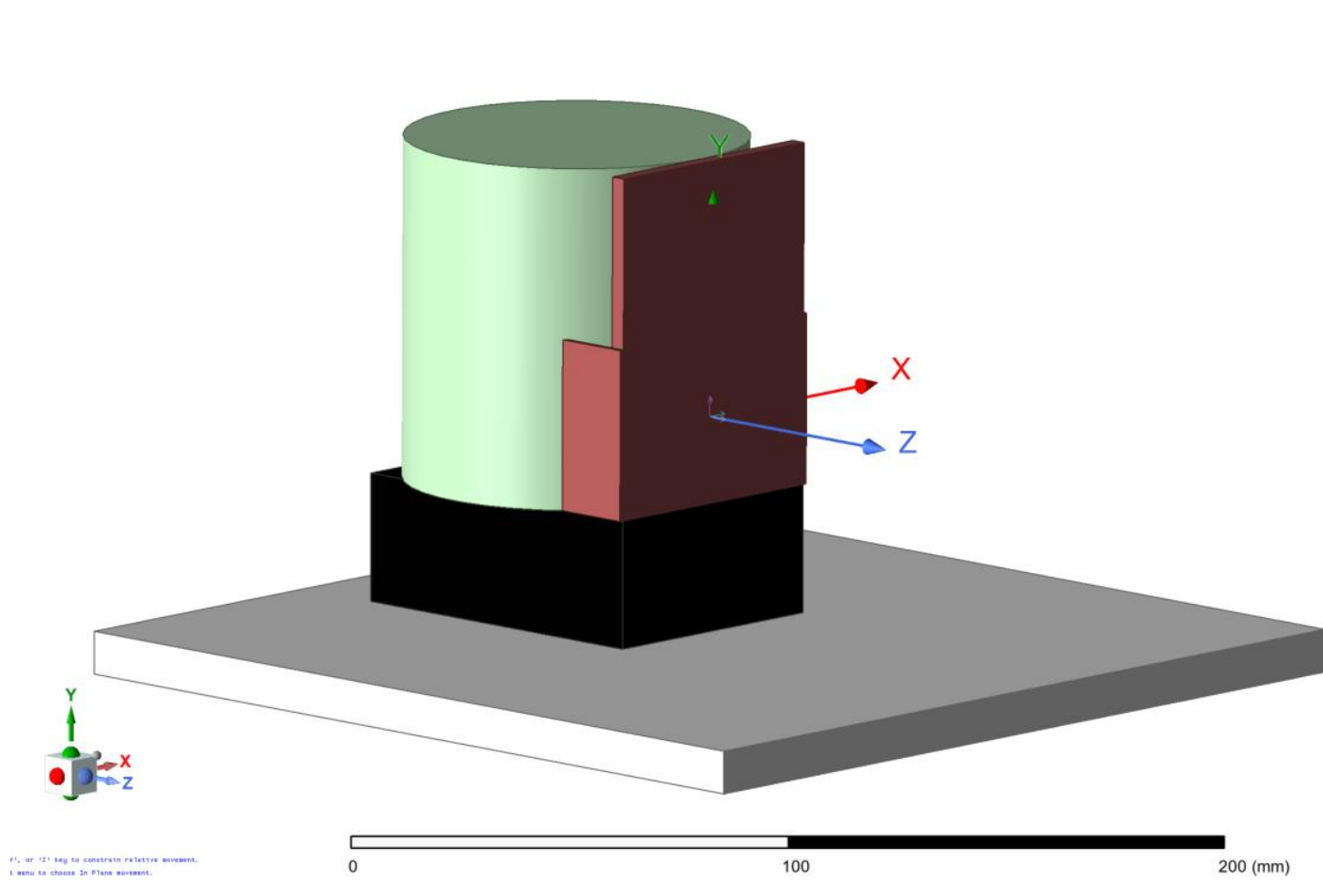


ANTENNA INSIDE A SHOE DEMO

MATERIALS

Standard RF libraries do not account for biological tissues or apparel. To accurately simulate the real-world dielectric loading, custom materials were defined based on empirical coaxial probe measurements from current literature:

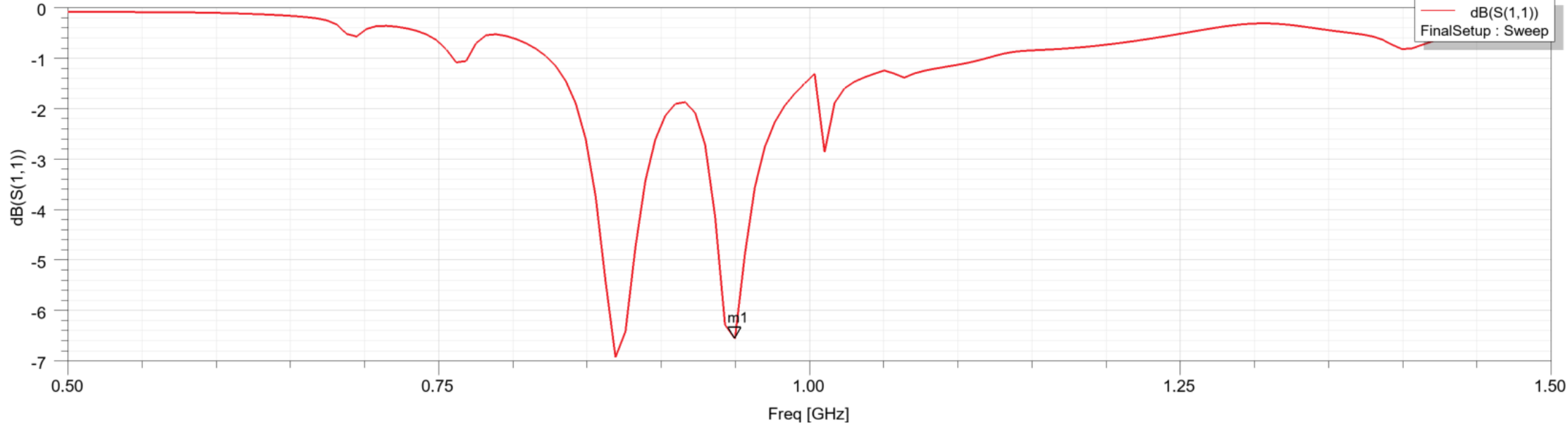
- Human Foot (Phantom Foot) –
 - Relative Permittivity - $\epsilon_{r_{foot}} \cong 40$
 - Loss - $\tan\delta = 0.015$
- Leather-
 - Relative Permittivity - $\epsilon_{r_{leather}} \cong 1.75$
 - Loss - $\tan\delta = 0.12$
- Sole (Rubber)
 - Relative Permittivity - $\epsilon_{r_{sole}} \cong 3.5$
 - Loss - $\tan\delta = 0.07$
- Floor (Concrete)- Data from ITU
 - Relative Permittivity - $\epsilon_{r_{concrete}} = 5.3$
 - Loss - $\tan\delta = 0.11$
 - Conductivity - $\sigma_{concrete} = 0.03 \frac{S}{m}$



ANTENNA MOCK MODEL

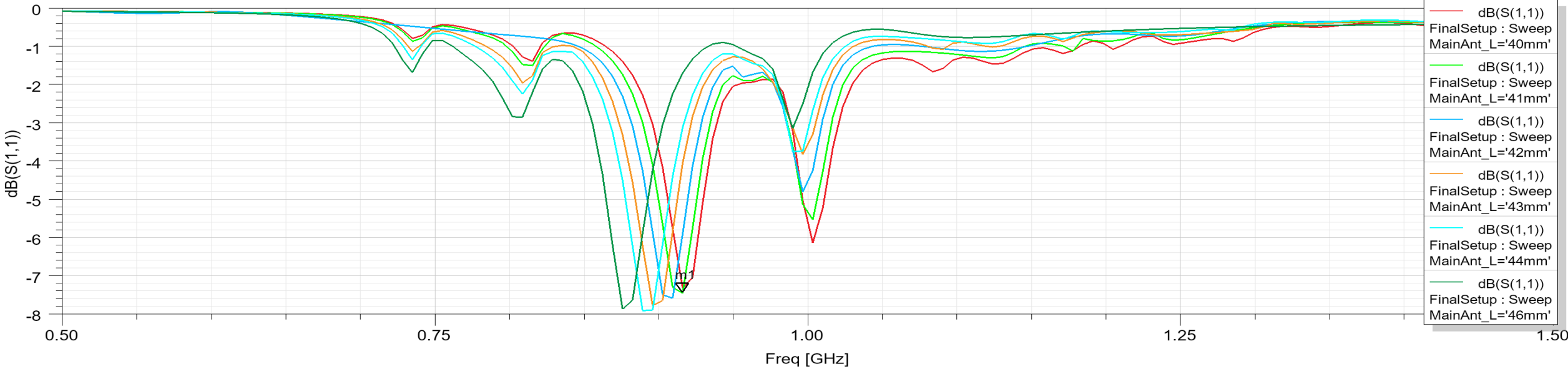
S Parameter Shoe Initial

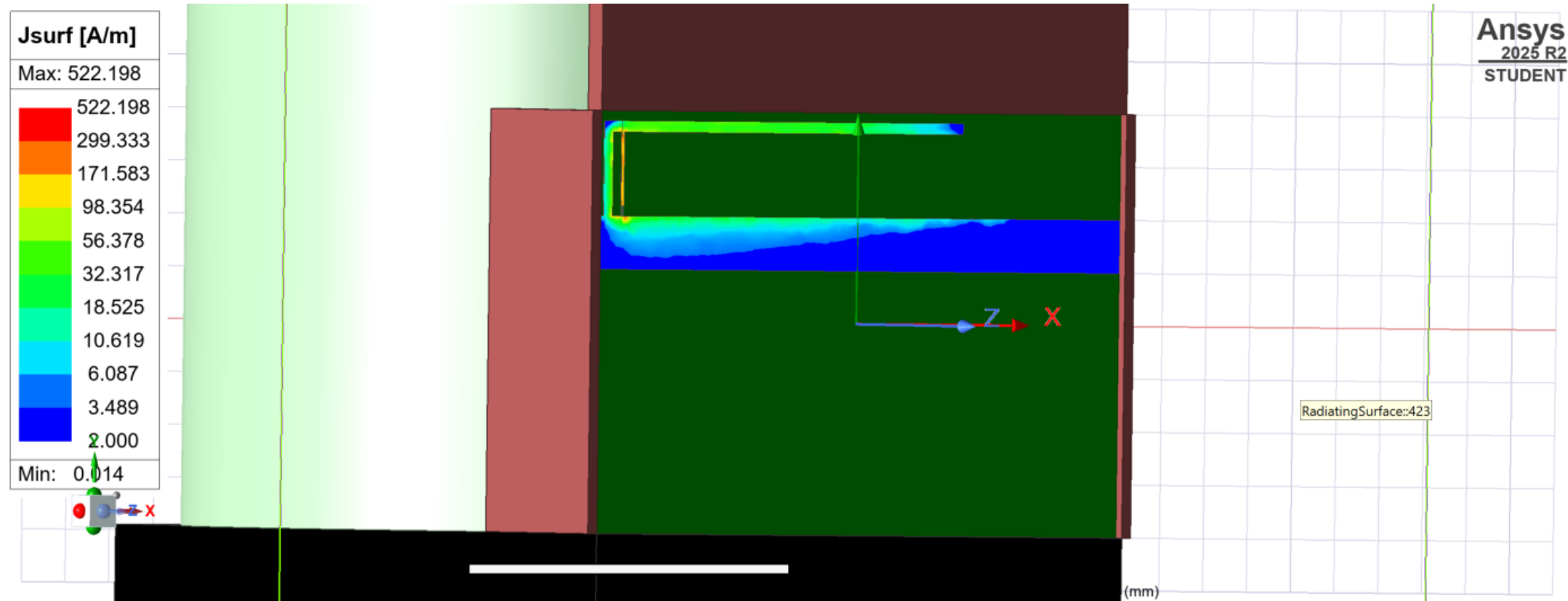
Name	X [GHz]	Y
m1	0.9497	-6.5558



S Parameter Ant L Optimizer

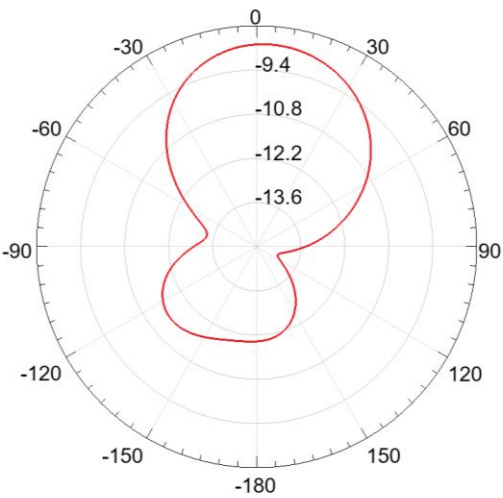
Name	X [GHz]	Y
m1	0.9161	-7.4499





SURFACE CURRENT PLOT

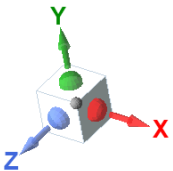
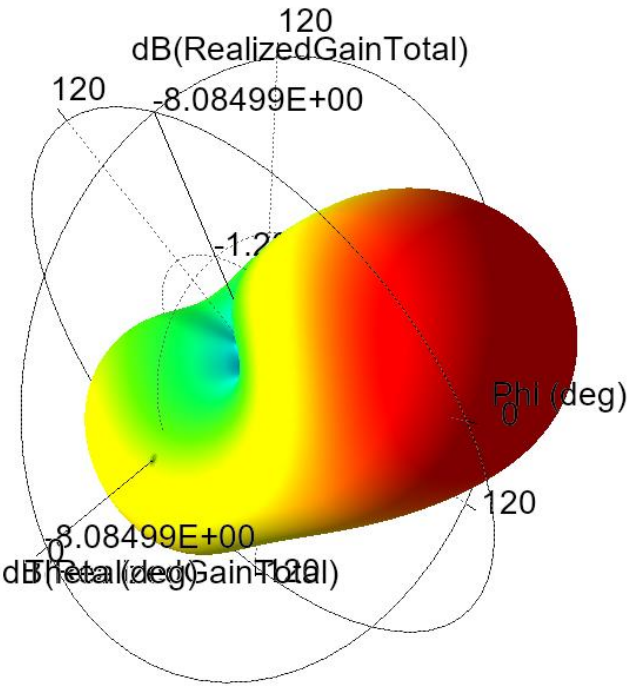
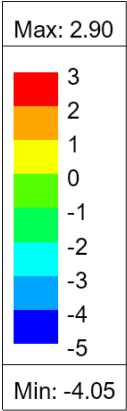
Realized Gain AZ Plot Shoe Ant



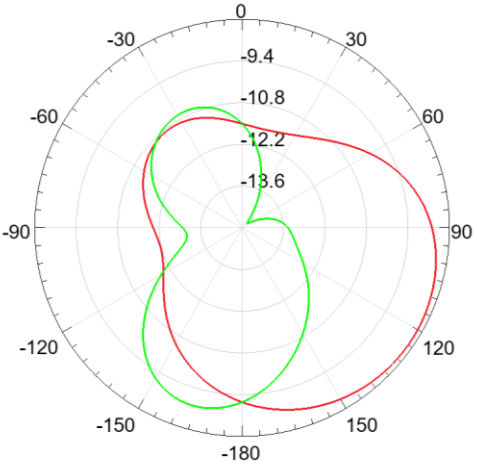
ShoeMockDesign1
Ansys
2025 R2
STUDENT

Realized Gain Show Ant

Ansyz Inc.

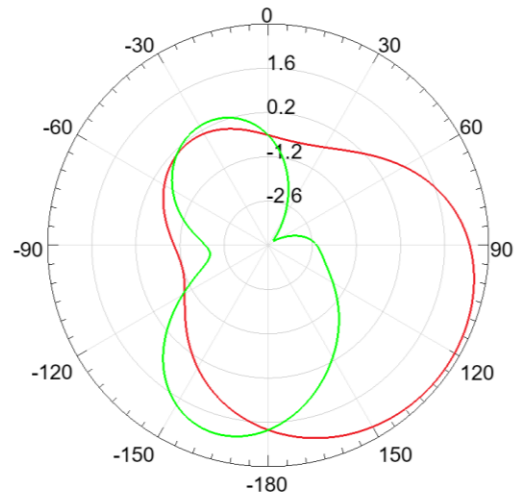


Realized Gain EL Plot Shoe Ant



ShoeMockDesign1
Ansys
2025 R2
STUDENT

Directivity EL Plot Shoe Ant

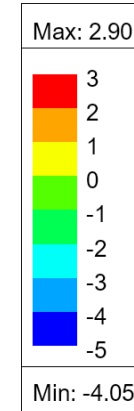


ShoeMockDesign1
Ansys
2025 R2
STUDENT

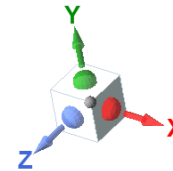
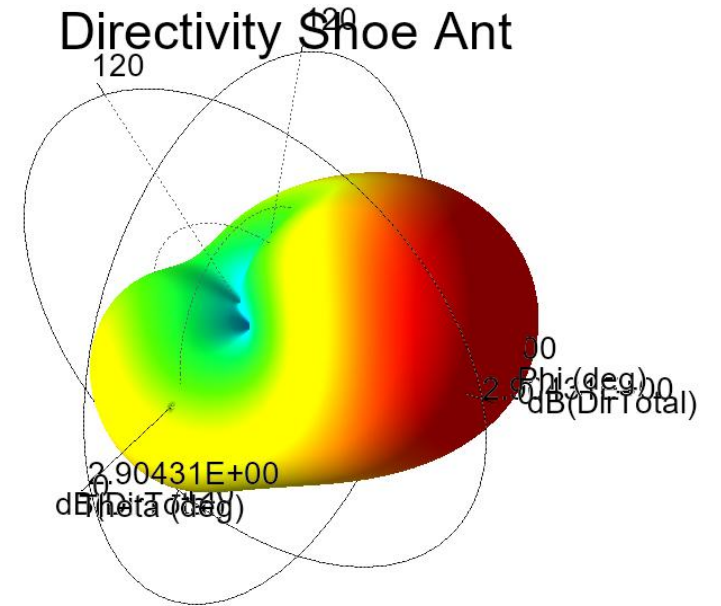
— dB(DirTotal)
FinalSetup : LastAdaptive
Freq='0.915GHz' Phi='0deg'

— dB(DirTotal)
FinalSetup : LastAdaptive
Freq='0.915GHz' Phi='90deg'

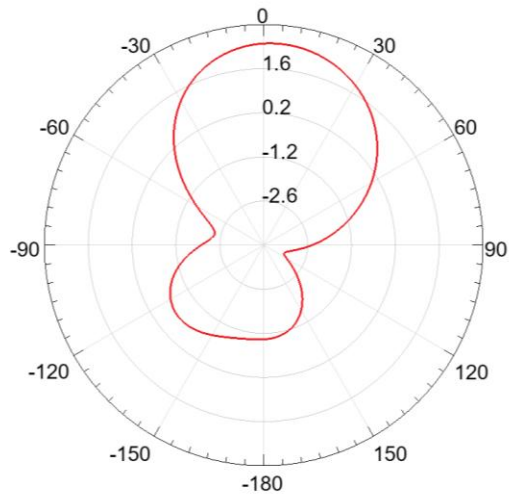
Ansys Inc.



Directivity Shoe Ant



Directivity AZ Plot Shoe Ant



ShoeMockDesign1
Ansys
2025 R2
STUDENT

— dB(DirTotal)
FinalSetup : LastAdaptive
Freq='0.915GHz' Theta='90deg'



THANK YOU FOR LISTENING